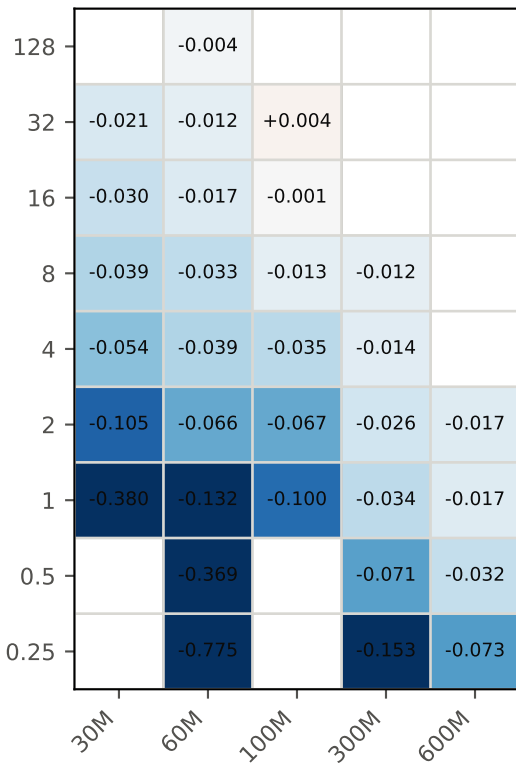
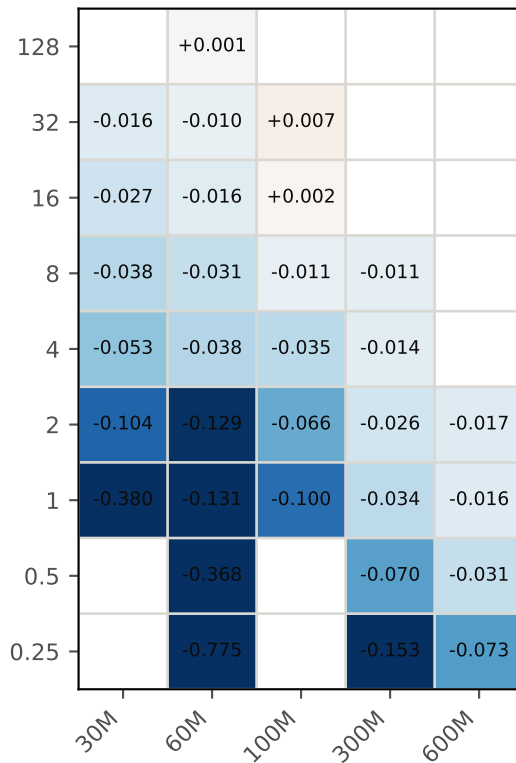


Robustness to Gaussian weight perturbation: muon vs adamw ($\gamma \leq 0.02$)

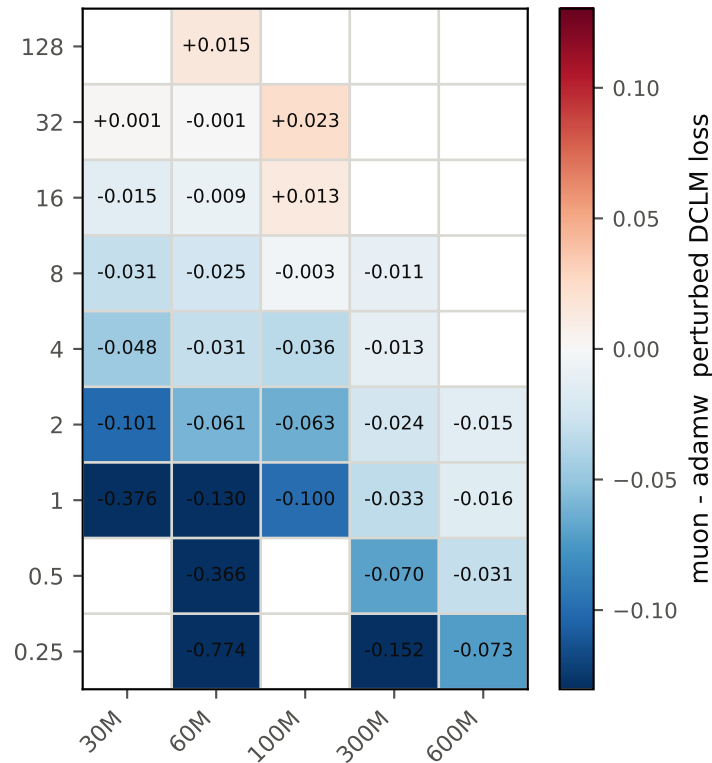
$\gamma = 0.005$



$\gamma = 0.01$



$\gamma = 0.02$



muon - adamw perturbed DCLM loss



cell = perturbed held-out DCLM loss, tuned muon base minus tuned adamw base | negative (blue) = muon degrades less | white = no data

colour scale clipped at ± 0.130 (85th pct); 14 cell(s) beyond it keep their printed value